1975

Sex Bias in Graduate Admissions: Data from Berkeley

Measuring bias is harder than is usually assumed, and the evidence is sometimes contrary to expectation.

P. J. Bickel, E. A. Hammel, J. W. O'Connell

Determining whether discrimination because of sex or ethnic identity is being practiced against persons seeking passage from one social status or locus to another is an important problem in our society today. It is legally important and morally important. It is also often quite difficult. This article is an exploration of some of the issues of measurement and assessment involved in one example of the general problem, by means of which we hope to shed some light on the difficulties. We will proceed in a straightforward and indeed naive way, even though we know how misleading an unsophisticated approach to the problem is. We do this because we think it quite likely that other persons interested in questions of bias might proceed in just the same way, and careful exposure of the mistakes in our discovery procedure may be instructive.

Data and Assumptions

The particular body of data chosen for examination here consists of applications for admission to graduate study at the University of California, Berkeley, for the fall 1973 quarter. In the admissions cycle for that quarter, the Graduate Division at Berkeley received approximately 15,000 applications, some of which were later withdrawn or transferred to a different proposed entry quarter by the applicants. Of the applications finally remaining for the fall 1973 cycle 12,763 were sufficiently complete to permit a decision to admit or to deny admission. The question we wish to pursue is whether the decision to admit or to deny was influenced by the sex of the applicant. We cannot know with any certainty the influences on the evaluators in the Graduate Admissions Office, or on the faculty reviewing committees, or on any other administrative personnel participating in the chain of actions that led to a decision on an individual application. We can, however, say that if the admissions decision and the sex of the applicant are statistically associated in the results of a series of applications, we may judge that bias existed, and we may then seek to find whether discrimination existed. By "bias" we mean here a pattern of association between a particular decision and a particular sex of applicant, of sufficient strength to make us confident that it is unlikely to be the result of chance alone. By "discrimination" we mean the exercise of decision influenced by the sex of the applicant when that is immaterial to the qualifications for entry.

The simplest approach (which we shall call approach A) is to examine the aggregate data for the campus. This approach would surely be taken by many persons interested in whether bias in admissions exists on any campus. Table 1 gives the data for all 12,763 applications to the 101 graduate departments and interdepartmental graduate majors to which application was made for fall 1973 (we shall refer to them all as departments). There were 8442 male applicants and 4321 female applicants. About 44 percent of the males and about 35 percent of the females were admitted. Just this

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Gender contributes to personal research funding success in The Netherlands

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We examined the application and review materials of three calls (n = 2,823) of a prestigious grant for personal research funding in a national full population of early career scientists awarded by the Netherlands Organization for Scientific Research (NWO). Results showed evidence of gender bias in application evaluations and success rates, as well as in language use in instructions and evaluation sheets. Male applicants received significantly more competitive "quality of researcher" evaluations (but not "quality of proposal" evaluations) and had significantly higher application success rates than female applicants. Gender disparities were most prevalent in scientific disciplines with the highest number of applications and with equal gender distribution among the applicants (i.e., life sciences and social sciences). Moreover, content analyses of the instructional and evaluation materials revealed the use of gendered language favoring male applicants. Overall, our data reveal a 4% "loss" of women during the grant review procedure, and illustrate the perpetuation of the funding gap, which contributes to the underrepresentation of women in academia.

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Women are still underrepresented in academia today. Despite various attempts to promote gender equality (e.g., affirmative action initiatives, quotas), female scientists are less likely to get offered tenure, are judged to be less competent, receive less payment and research facilities, and are less likely to be awarded research grants compared with male scientists (1–3). Over time, this type of bias accumulates and contributes to the attrition of women from associated with male traits and considered necessary for academic career success (20). Moreover, women still earn on average 18% less than their male colleagues for the same work with similar responsibilities (3). Although the salary gap seems to narrow for early career researchers, women in top academic positions are still substantially underpaid compared with men. Finally, across different career phases, success rates for female scientists applying for research funding tend to be lower than for male scientists (3, 21, 22). Even when overall success rates for men and women are equal, women receive less research funding than men, and are less often listed as principal investigators (23–25). Closing the funding gap is of particular importance, because this may help retain women in academia and foster the closing of other gaps by facilitating negotiations about salaries, research facilities, and promotion opportunities.

Current Study

To investigate the possibility of a funding gap, we examined a national full population of early career researchers who applied for a prestigious personal grant between 2010 and 2012 (Innovational Research Incentives Scheme Veni; n = 2,823, with 42.1% female applicants) awarded by the Netherlands Organization for Scientific Research (NWO). The NWO made available anonymized data from their archives for the purpose of this study, and approved publication of this research.

Our focus in this study was twofold. First, we tested for applicant gender differences in success rates and application evaluations. In

Dutch research funding, gender bias, and Simpson's paradox

Based on, among other criteria, three consecutive years of grant applications to the "Veni programme" of the Netherlands Organization for Scientific Research (NWO), van der Lee and Ellemers (1) conclude that these data "provide compelling evidence of gender bias in personal grant applications to obtain research funding." One of the main results this claim is based upon is that of the 1,635 applications by males, 17.7% were successful, whereas of the 1,188 applications by females, only 14.4% were successful. When applying the χ^2 test to the data, the authors found a P value of 0.045 (1). This conclusion is based on the application of an inappropriate statistical procedure, and is therefore questionable, due to the so-called "Simpson's paradox."

Simpson's paradox dates back to Simpson (2) and gained familiarity after a report by Bickel et al. (3). In essence, Simpson's paradox states that an apparently significant relation between two variables in a contingency table can be due to a joint dependency on a third variable.

In table S1 of the article by van der Lee and Ellemers (1), a breakdown of grant applications per research discipline is presented. The proportion of female applicants varies from 11.8% visualizes these data and immediately shows a clear negative relation between the proportion of female applicants and the total success rate. It turns out that women tend to apply more often to competitive disciplines, such as health sciences and social sciences, whereas men apply more often to less competitive disciplines, such as physics and chemical sciences. In four of the nine disciplines, women have a higher success rate than men, and in five of the nine disciplines, men have a higher success rate than women. When taking into account that multiple comparisons are performed, the gender bias is significant (at the $\alpha = 0.05$ level) for none of the disciplines. Thus, when taking into account the spurious correlation, the statistical significance of the relation vanishes.

As a consequence, the conclusion of van der Lee and Ellemers (1) could be severely influenced by this issue, at least for the (large) part where this claim is based on the supposed P value of 0.045. The authors perform other tests on various aspects of the proposal review process, but the data provided in the paper are insufficient to check whether these tests suffer from Simpson's paradox as well. Because these tests study related issues, it is not unlikely that Dutch science funding. This scrutiny includes studying data from other years and programs. Interestingly, on the aggregate level, men received relatively more grants than women in eight of the 14 Veni calls, with the difference, again, not being statistically significant (4).

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